

Computer Science & IT

Database Management System



Query Languages

Lecture No. 07



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Recap of Previous Lecture

- ✓ Topic Aggregate functions
- ✓ Topic SQL commands
- ✓ Topic SQL clauses



Topics to be Covered



Topic

SQL clauses



Topic

Introduction to nested query



Topic

IN, ANY, ALL and EXISTS operators





Topic : SQL clauses

- ✓ **FROM** - The FROM clause in SQL is used to select the database tables.
- ✓ **WHERE** - The WHERE clause in SQL is used to select the tuples from the database table based on conditions specified with WHERE clause. { If where clause is not present, then all the tuples will be selected }
- ✓ **GROUP BY** - GROUP BY clause is used to group the result of WHERE clause based on attributes specified with GROUP BY clause. { If where clause is not present then group by clause will be applied on all tuples } { If Group By clause is not present then all tuples together will be treated as a single group }
- ✓ **HAVING** - HAVING clause can be used with or without GROUP BY clause. If group by clause is present, then condition specified with having clause will be used to select the groups that satisfy the specified condition. { Having Condⁿ is applied on each group }
- ✓ **ORDER BY** - The ORDER BY clause in SQL is used for sorting the records of the database based on attributes specified with ORDER BY clause.



Topic : Order of execution



Order of Execution:-

- ✓ 1. From
- ✓ 2. Where
- ✓ 3. Group By
- ✓ 4. Having
- ✓ 5. Select
⑤' distinct
- ✓ 6. Order BY



Topic : SQL clauses

✓ **FROM:-** From clause is used to select the tables from the database.

Student

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC

Select *

From Student

o/p of this query
will be complete Student
table.

Everything i.e. Select all attributes



Topic : SQL clauses



✓ **FROM:-** From clause is used to select the tables from the database.

Student

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S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC

Select Sid
From Student

⇒ o/p :-

Sid
S₁
S₂
S₃
S₄
S₅
S₆

Select distinct Sid
From Student



Topic : SQL clauses



✓ **FROM:-** From clause is used to select the tables from the database.

Student

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC

Select Sname $\Rightarrow$ o/p
From Student

Sname
A
A
B
A
C
C

Select distinct Sname
From Student

$\Rightarrow$ o/p =

Sname
A
B
C



Topic : SQL clauses



WHERE:- Used to select the tuples from the database table based on conditions specified with WHERE clause.

If where clause is not present
all tuple will be selected

Student

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC

★ Retrieve tuple from Student table
in which Marks are more than 40

③ Select *

① From Student

② Where Marks > 40

c/p=



Topic : SQL clauses



WHERE:- Used to select the tuples from the database table based on conditions specified with WHERE clause.

If where clause is not present
all tuple will be selected

Student

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC

★ Retrieve Sids of students from Student table where marks are more than or equal to 20 and less than or equal to 40

O/p=

Select Sid
from Student

Where (Marks ≥ 20 AND Marks ≤ 40)

O/p=

Sid
S1
S2
S5



Topic : SQL clauses



WHERE:- Used to select the tuples from the database table based on conditions specified with WHERE clause.

If where clause is not present
all tuple will be selected

Student

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC

★ Retrieve Sids of students from Student table where marks are more than or equal to 20 and less than or equal to 40

Select Sid
from Student
Where Marks Between 20 AND 40

Both are included
with Between

O/p =

Sid
S1
S2
S5



Topic : SQL clauses



✓ **GROUP BY:-** GROUP BY clause is used to group the result of WHERE clause. { If Where clause is not present all the tuples of the table are grouped based on attributes specified }

Query: Retrieve names of all branches along with maximum marks in that branch.

H.W.:-

STUDENT

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC

➤ Retrieve maximum marks from student table

Select maxc (Marks)
From Student

q/p:

60

➤ Retrieve names of all the branches from Student table

Select distinct Branch
From Student

q/p=

Branch
CS
IT
EC

- Retrieve names of all the branches along with maximum marks in those branches.

Select Branch, max(Marks)
From Student

Aggregate function + Attribute from table

invalid query

In select clause we can not specify any attribute along with aggregate function until those attributes are present in group by clause



Topic : SQL clauses



NOTE:-

1. We can not select any attribute in SELECT clause along with aggregate function if those attributes are not present in GROUP BY clause.
2. If aggregate function is used along with GROUP BY clause, then aggregate function is applied on each group.



Topic : SQL clauses

GROUP BY:- GROUP BY clause is used to group the result of WHERE clause.

Query: Retrieve names of all branches along with maximum marks in that branch.

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC

- Select Branch, Max(Marks)
① From Student
② Group by (Branch)

Q/P

Branch	Max(Marks)
CS	60
IT	40
EC	60



Topic : SQL clauses



HAVING:- HAVING condition is applied on each group.

{ Provided group by clause is present }

Query: Retrieve branch names with average marks more than or equal to 40.

- ④ Select Branch
- ① From Student
 - ② Group By (Branch)
 - ③ Having (Avg(Marks) $\geq$ 40)

Having Condⁿ is used to select the groups that are satisfying the specified condition

Student

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC

50

30

60

yp Branch
CS
EC



Topic : SQL clauses



NOTE:-

1. WHERE condition is applied on each tuple whereas HAVING condition is applied on each group.
2. We can use HAVING condition without GROUP BY clause as well, but in that case all the tuples together are treated as a single group.



Topic : SQL clauses



ORDER BY:- This clause is used to sort the result in ascending or descending order based on values of attribute specified with ORDER BY clause.

By default order is ascending order.

Otherwise $\left\{ \begin{array}{l} \text{For ascending order we use Asc} \\ \text{For descending order we use "desc"} \end{array} \right.$

Student

Sid	Sname	Marks	Branch
S1	A	40	CS
S2	A	20	IT
S3	B	60	CS
S4	A	60	EC
S5	C	40	IT
S6	C	NULL	EC

→ Select Sid, Sname
From Student
Where (Marks $\geq$ 40) $\Rightarrow$ o/p =

Sid	Sname
S ₁	A
S ₂	B
S ₄	A
S ₅	C

→ Select Sid, Sname
From Student $\Rightarrow$ o/p
Where (Marks $\geq$ 40)
Order by (Sname)

Sid	Sname
S ₁	A
S ₄	A
S ₂	B
S ₅	C

By default ascending
We may write "Asc" also

→ Select Sid, Sname
From Student $\Rightarrow$ o/p
Where (Marks $\geq$ 40)
Order by (Sname) desc

Sid	Sname
S ₅	C
S ₂	B
S ₄	A
S ₁	A

for descending
order

Query:- From the above Student table retrieve Sids of students who scored maximum Marks.

* Select Sid From Student Having (Max(Marks))
it is not a Condⁿ

* Select Sid From Student Where (Marks = Max(Marks))

Invalid: We can not use aggregate function directly within Where clause

→ Select Sid From Student Having (Marks = Max(Marks))

* Select Sid From Student Where (Max(Marks)) Not a Condⁿ

All of them are Wrong.

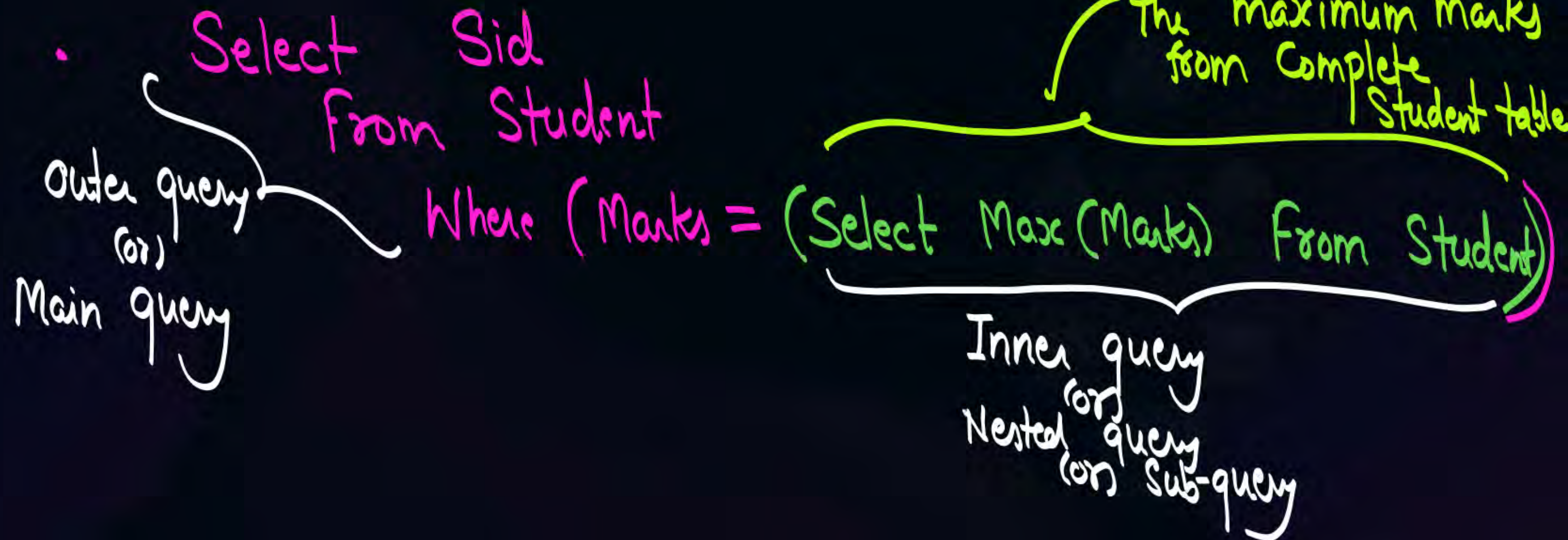


Topic : Nested queries

Sub-query Concept :-



- Retrieve Sids of all students from Student table who scored maximum marks.



Nested Query (Sub-query)

Independent Nested query
(Independent Sub-query)

Correlated Nested query
(Correlated Sub-query)

- If inner query can be executed independently, then it is called independent nested query

- * When execution of inner query requires the value of an attribute from the relation specified in outer query, then it is called Correlated Nested query

Nested Query (Sub-query)

Independent Nested query
(Independent Sub-query)

Correlated Nested query
(Correlated Sub-query)

eg:

Select S.Sid
From Student S
Where (S.Marks = (Select Max(Marks)
From Student))

it can be executed
independently
∴ Independent Nested
query

eg: Select R.A
From R

Where (Select *
From S
Where (S.B = R.C))

∴ it is
called
Correlated
query

it can not be executed
independently because we need
the value of an attribute from
relation used in outer query

* "Order of Execution w.r.t. independent Nested query".

① Inner query will be executed first and it will produce its output.

and then ② Outer query will execute, it will use the o/p produced by inner query, for its execution.



Topic : Independent nested query

Select

Sid

}

finally

those Sids
will be produces
as o/p by
main query

=> o/p =

Sid
S3
S4

From

Student

Where

(Marks =

(Select Max(Marks)
From Student)

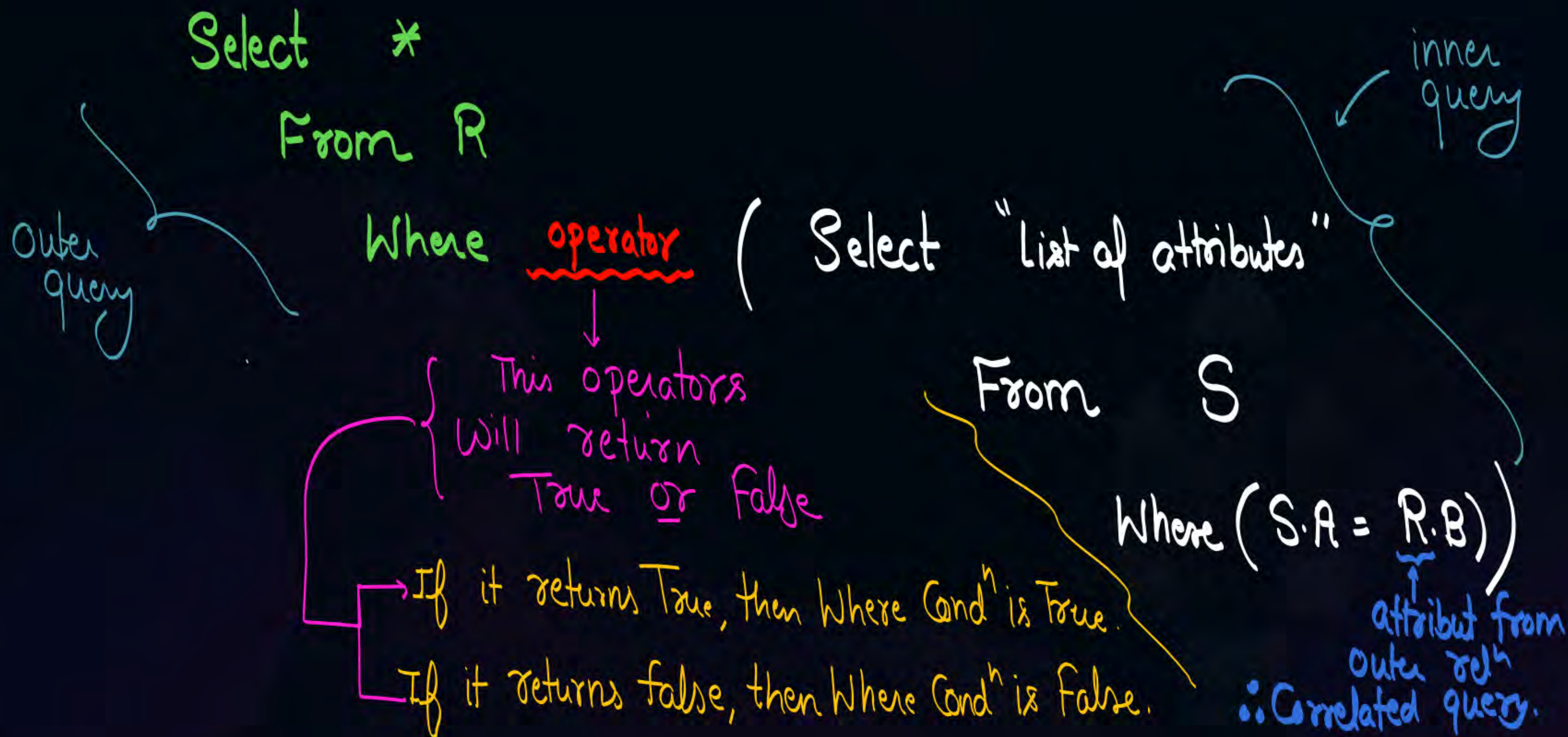
inner query will execute
first, and it will produce

o/p = 60

When outer query
executes w.o. o/p produced
by inner query, then
tuples w.o. Sid = S1 and
Sid = S2 are selected



Topic : Correlated nested query



HoNo:-

Read about,

IN, ANY, ALL and EXISTS Operators.



Topic : Correlated nested query

Outer query

Select *

From R

Where operator (

inner query

Select "List of attributes"

From S

Where (S.A = R.B)

↑
attribute from
outer relⁿ
∴ Correlated query.



2 mins Summary



Topic

SQL clauses

Topic

Introduction to nested query

Topic

IN, ANY, ALL and EXISTS operators

THANK - YOU